

Report Link:

[Document.docx](#)

Note: Undergraduate students will work in groups of three and complete Tasks 1-7.

Graduate students will work in pairs and complete Tasks 1-8.

See the file attached above for the project requirements.

Overview

In this project, you will design an analog integrate-and-fire (I&F) neuron circuit and use it to inform the behavior of a spiking neural network (SNN). You will bridge circuit-level design and SNNs by embedding your circuit's characteristics into a Python-based SNN and validating consistency between circuit simulations and software models. This project requires you to design and characterize a transistor-level neuron circuit, extract its behavior and reproduce it in software, use the circuit-informed neuron in an SNN, validate consistency between circuit and software, and evaluate system-level performance under realistic neuron constraints.

Report Requirements (20 points)

(Report submission deadline: Monday May 4th, 11:59PM.)

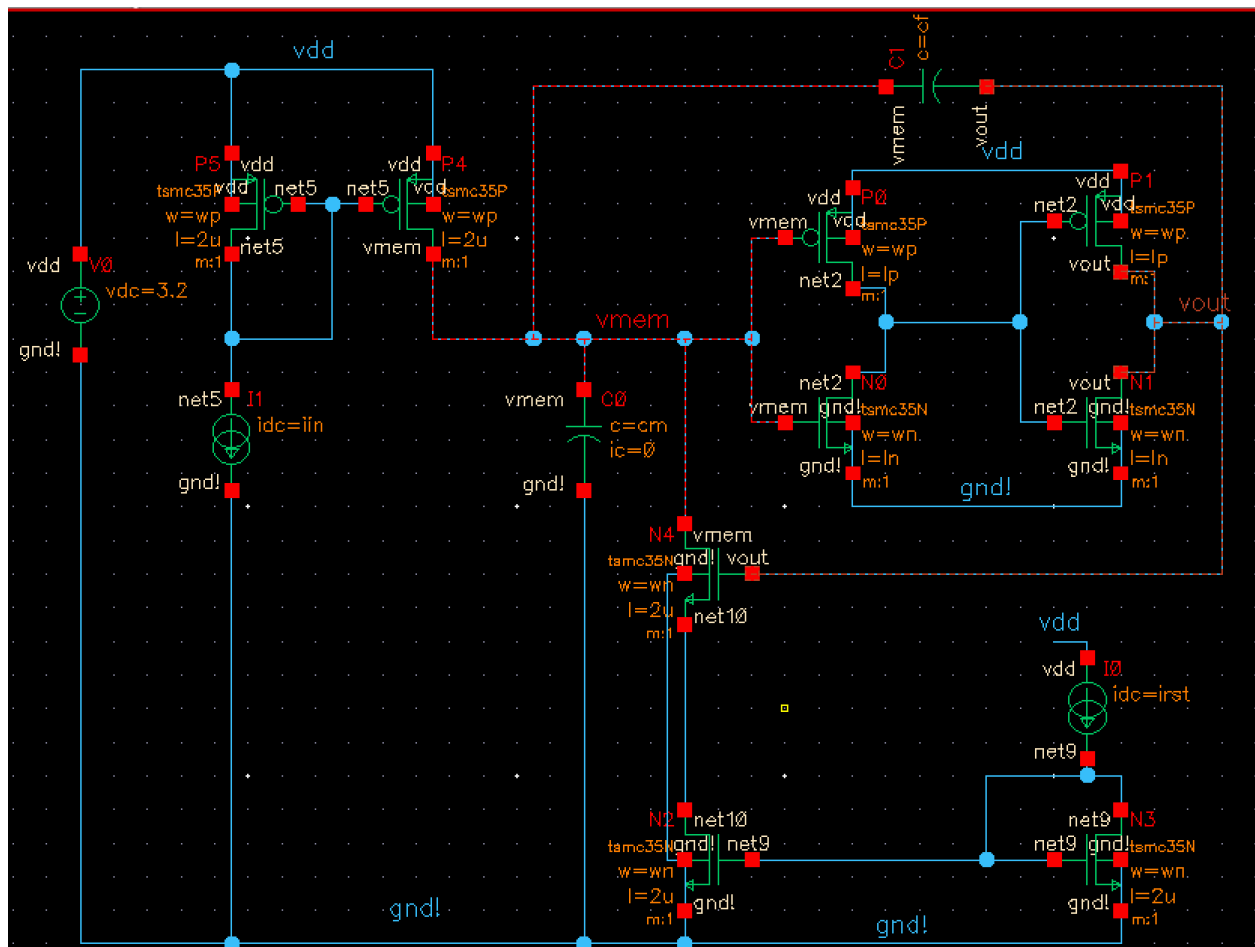
The final report must be clear with plots and tables in the report, and codes in the appendix. Follow the following report structure using the IEEE Conference Template available: <https://www.ieee.org/conferences/publishing/templates>

[Links to an external site.](#)

Page limit: 6 pages for all of the sections below.

1. [Introduction](#)
2. [Methodology](#)
3. [Results](#)
4. [Conclusion](#)
5. [Appendix](#)
6. [References](#)
7. [Author Contributions](#): Each student must provide a brief description of their contributions (approximately two sentences per student).

Part 1A and B:



Design Variables

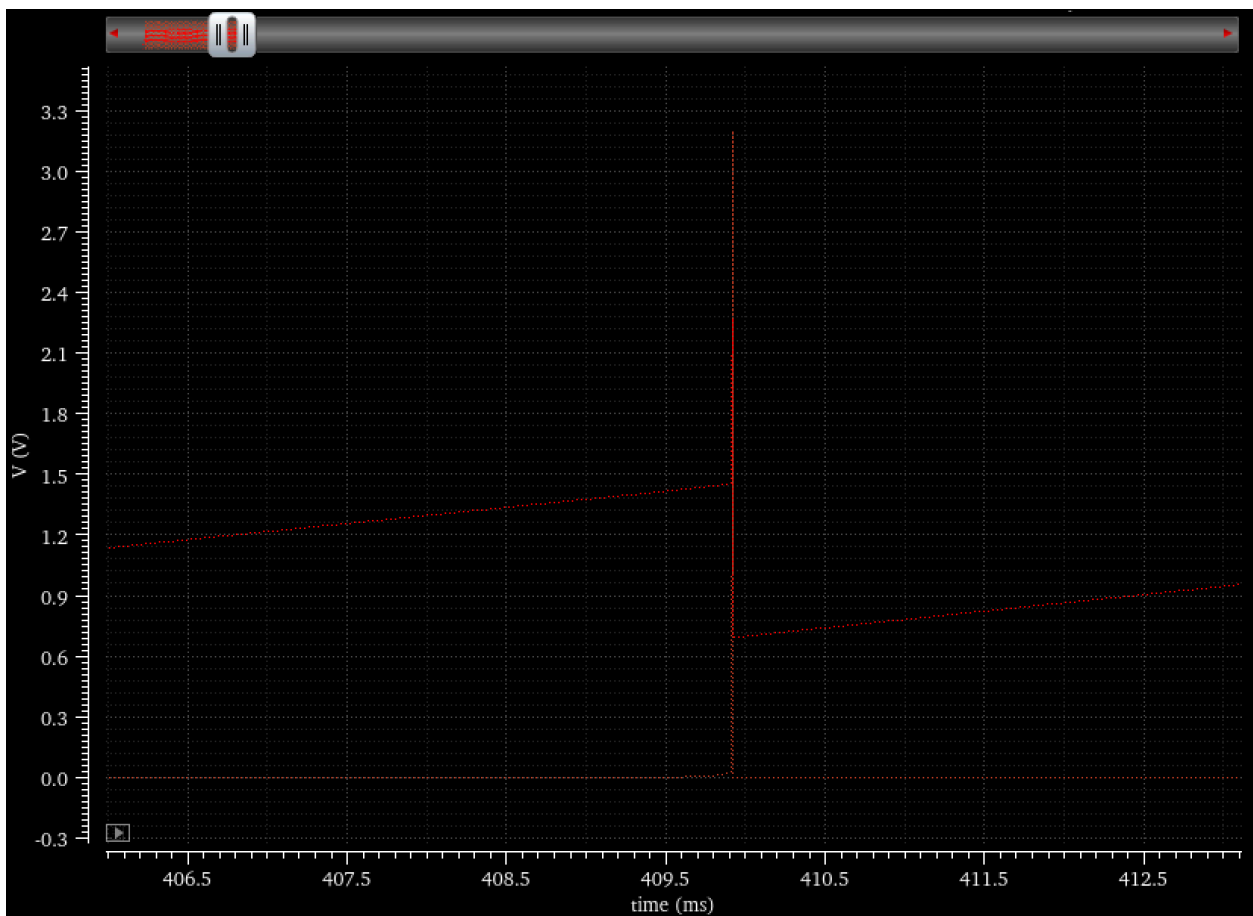
	Name	Value
1	cfb	3.557p
2	cm	10p
3	iin	1n
4	ln	600n
5	lp	600n
6	irst	2u
7	wn	10u
8	wp	20u

Analyses

Type	Enable	Arguments
1 tran	☒	0 500m

Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 vout		☒	☐	allv
2 vmem		☒	☐	allv



E9	frequency(VT("/vout"))	107.0
E10	frequency(clip(VT("/vout") 0.2 0.5))	107.0

Design Variables

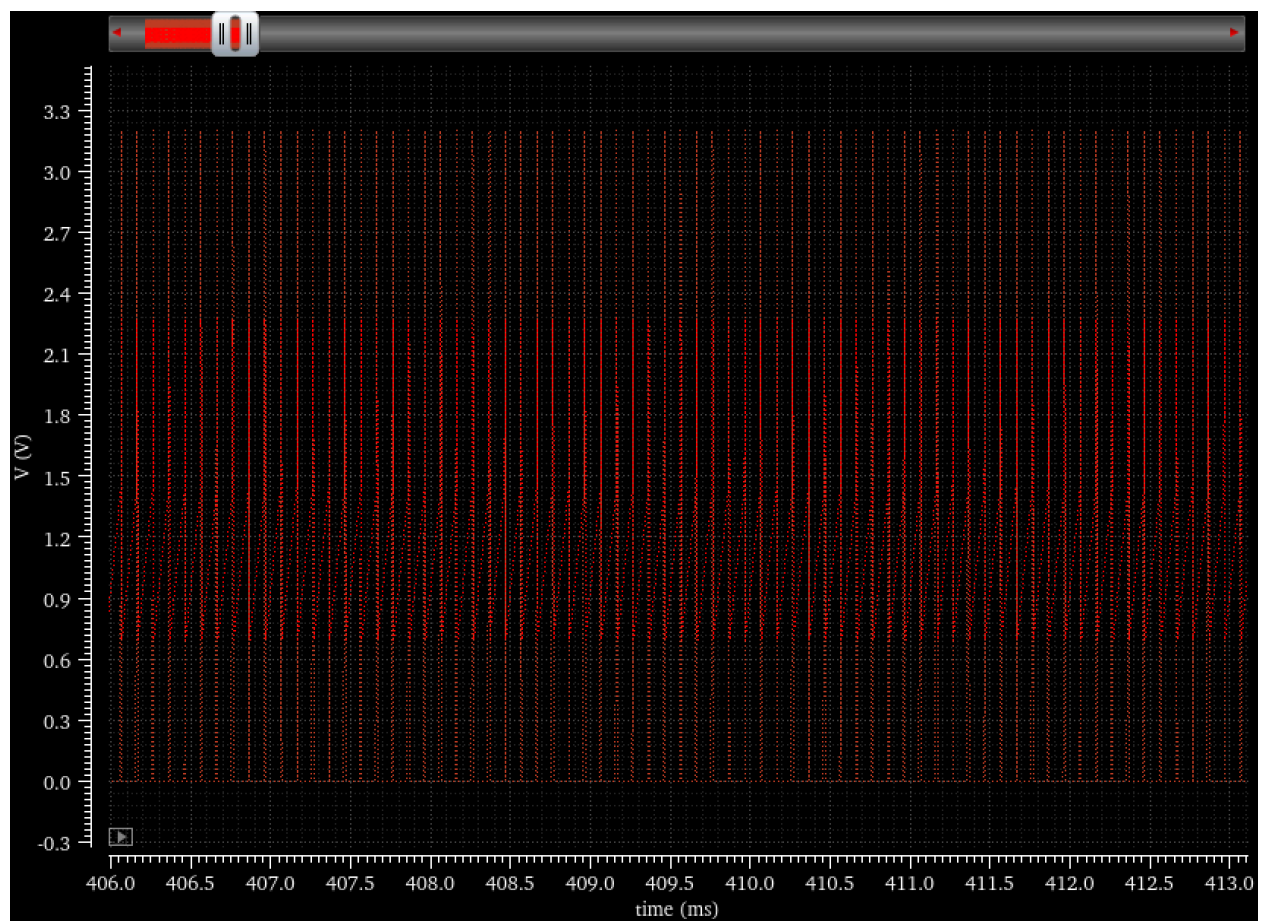
	Name	Value
1	cfb	3.557p
2	cm	10p
3	iin	100n
4	ln	600n
5	lp	600n
6	irst	2u
7	wn	10u
8	wp	20u

Analyses

Type	Enable	Arguments
1 tran	☒	0 500m

Outputs

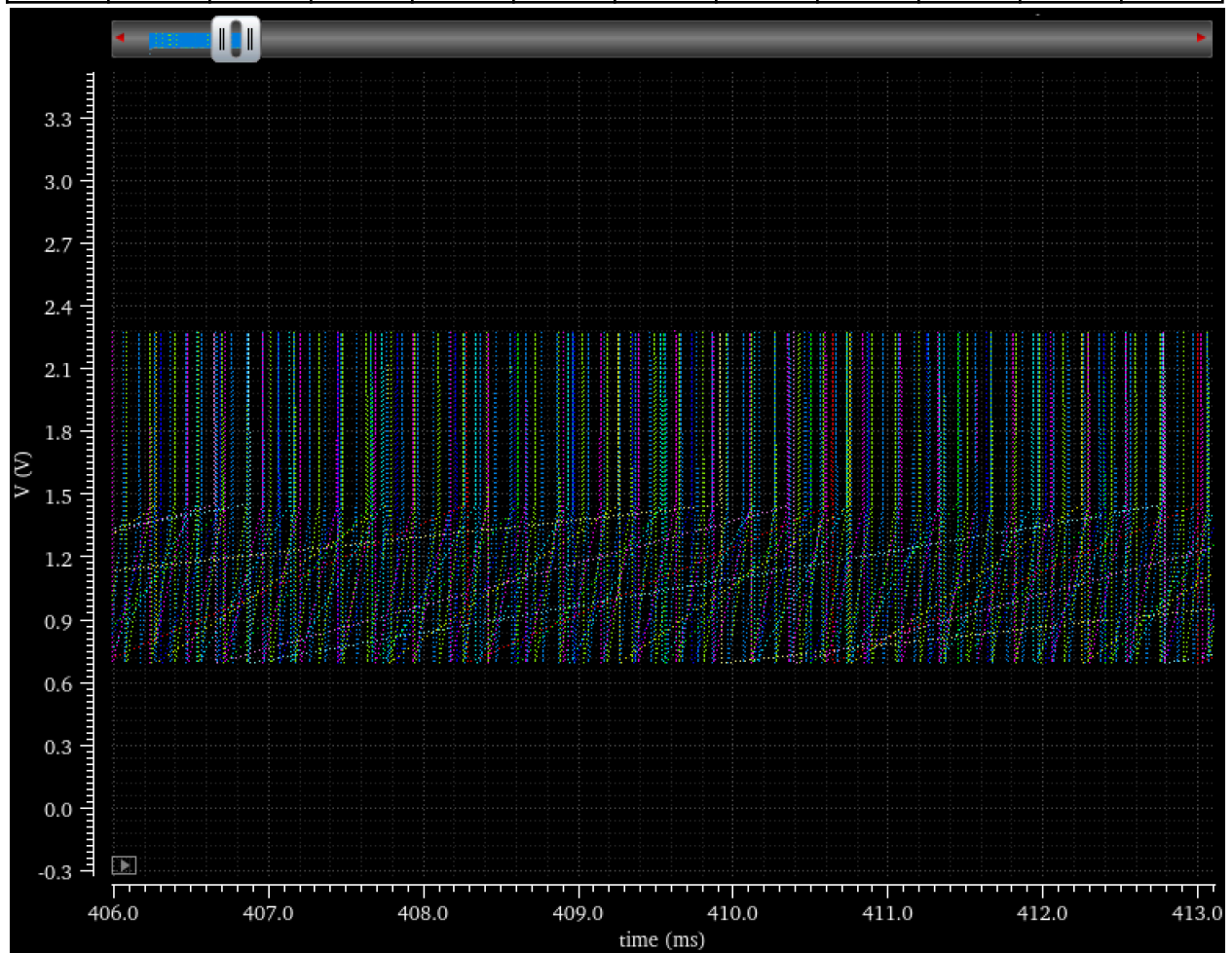
Name/Signal/Expr	Value	Plot	Save	Save Options
1 vout		☒	☐	allv
2 vmem		☒	☐	allv



E9	frequency(VT("/vout"))	9.998E3
E10	frequency(clip(VT("/vout") 0.2 0.5))	9.998E3

1C:

Input	1nA	10nA	20nA	30nA	40nA	50nA	60nA	70nA	80nA	90nA	100nA
Hz	107 Hz	1.057 kHz	2.098 kHz	3.126 kHz	4.142 kHz	5.146 kHz	6.139 kHz	7.12 kHz	8.091 kHz	1.05 kHz	9.998 kHz



Link to dataset of sweep shown above:

https://drive.google.com/file/d/1uzXwGMCM8-TPSR3ekRx3UXGgEsqhILKf/view?usp=drive_link

3.

Design Variables

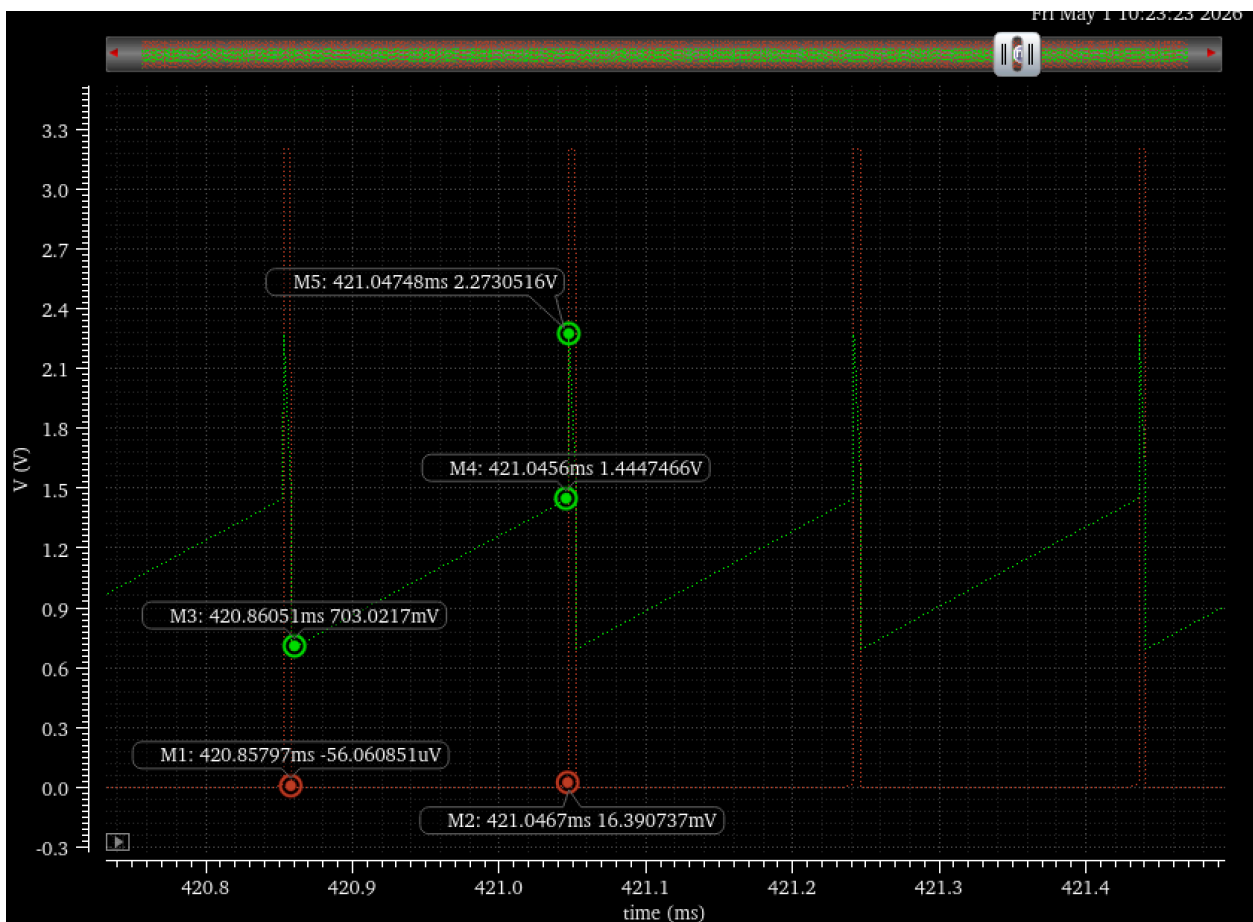
	Name	Value
1	b	3.557p
2	m	10p
3	n	50n
4		600n
5		600n
6	st	2u
7	n	10u
8	p	20u

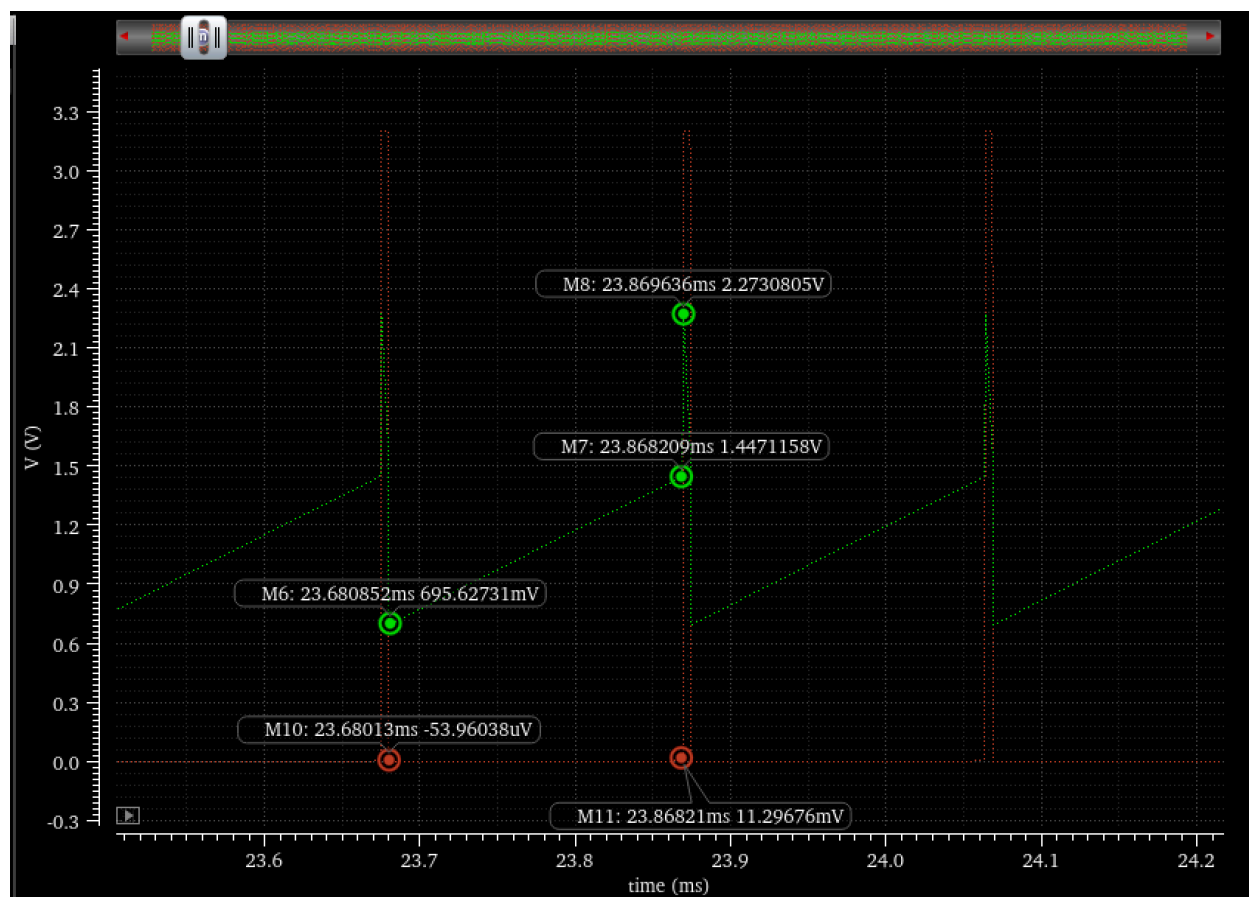
Analyses

	Type	Enable	Arguments
1	tran	☒	0 500m

Outputs

	Name/Signal/Expr	Value	Plot	Save	Save Options
1	vout		☒	☐	allv
2	vmem		☒	☐	allv





AC phase	
Current 1	0 A
Current 2	iin A
Delay time	
Rise time	100n s
Fall time	100u s
Pulse width	
Period	

